Packet 1 and 2

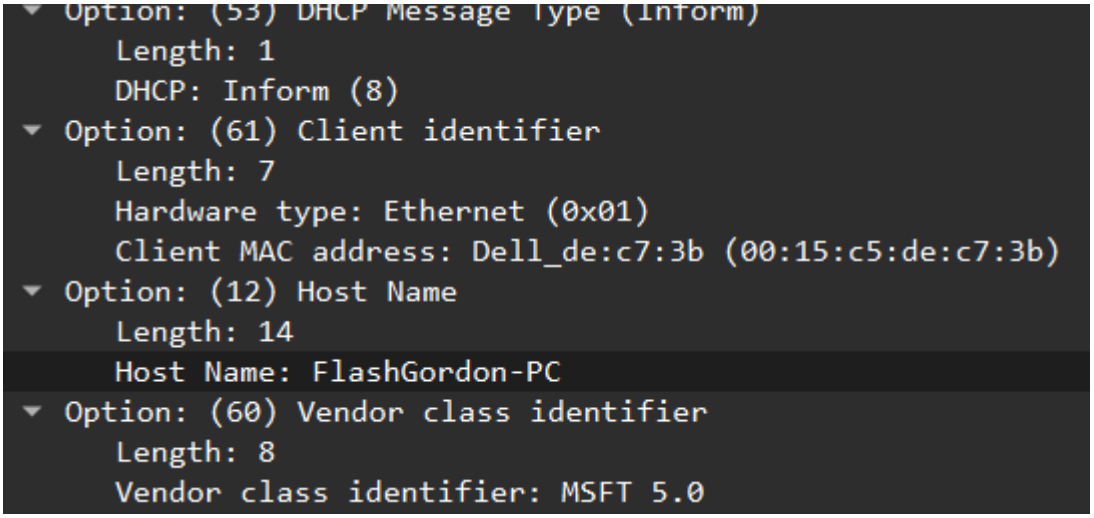
DNS request and response, 192.168.1.96 requests IP for "matied.com", "Cisco_65:3b:c1" responds with the IP "119.28.70.207"

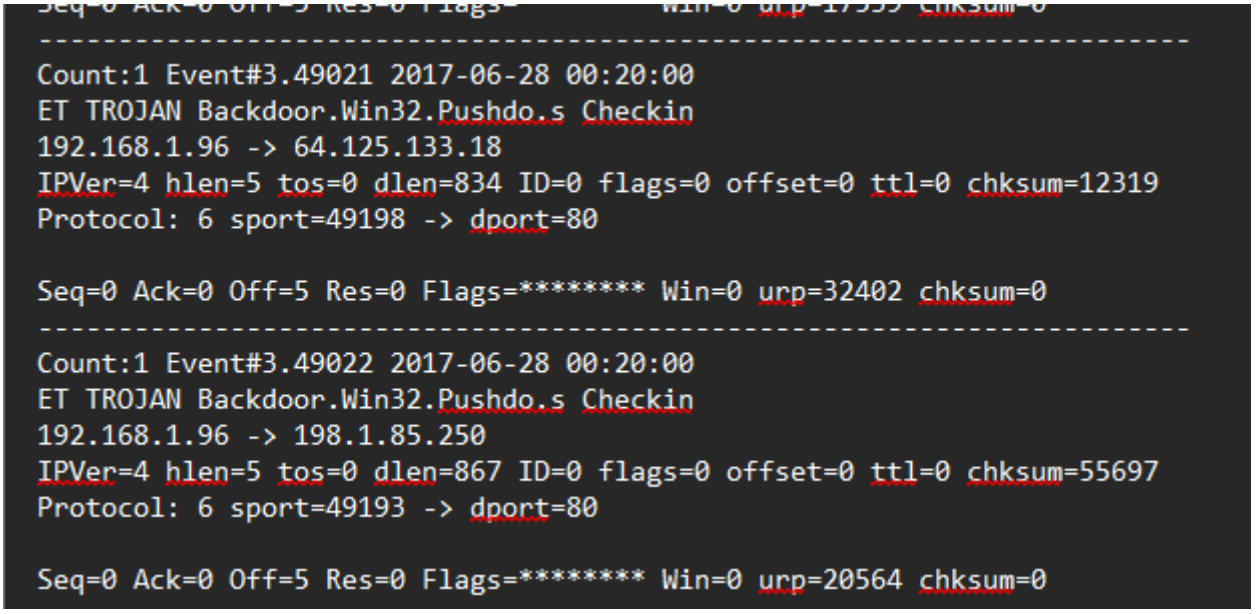


Used the tree to find the IP and mac

To find the http Host names, I added a custom column with “http.host” and filter the packets by http.request - this shows all the requests, and they are all coming from the same infected PC

To find host name, used the initial DHCP Host Name Option for the ip addr

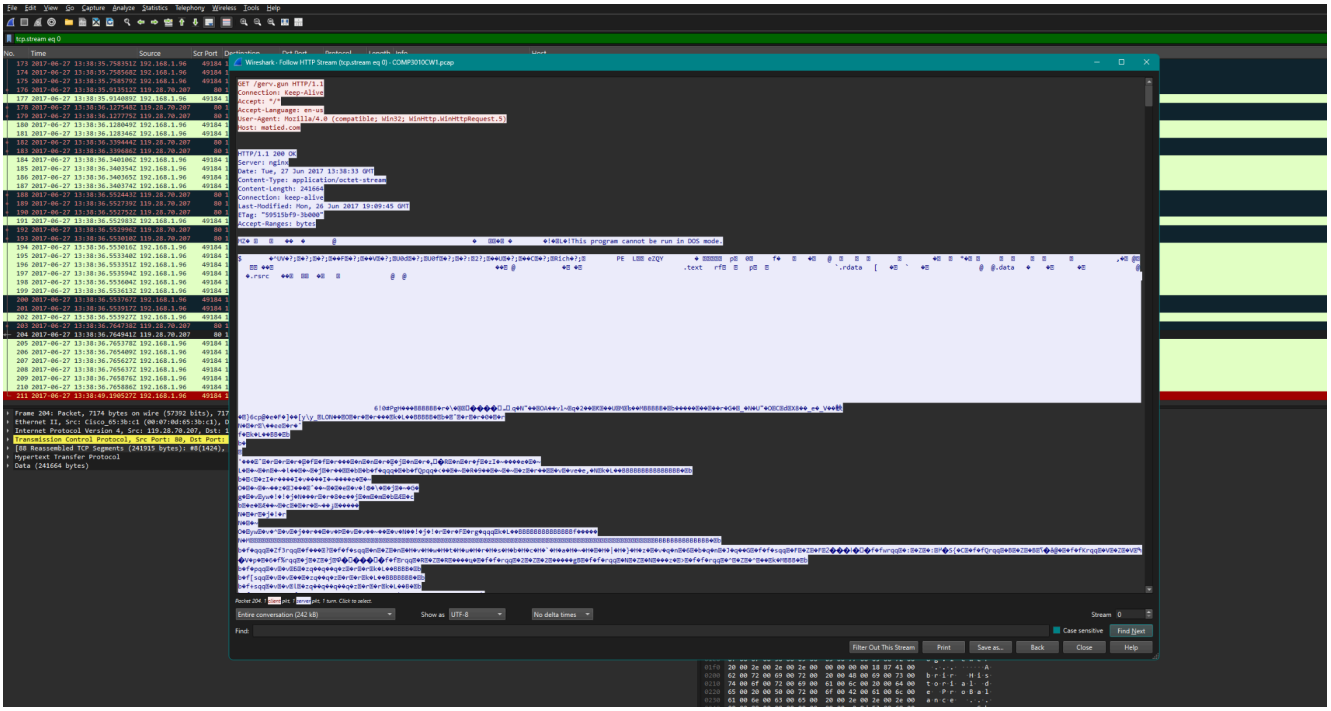




^

Attack is a pushdo variant

Exported gerv.gun as a http object, then used Get-FileHash to find the has of gerv.gun



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\jake> Get-FileHash "C:\Users\jake\OneDrive\Documents\COMP_3010_local\gerv.gun"

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          0931537889C35226D00ED26962ECACB140521394279EB2ADE7E9D2AFCF1A7272       C:\Users\jake\OneDrive\Docume...


PS C:\Users\jake>
```

Then Did the same for the two exe files trow.exe and wp.exe

```
yright (C) Microsoft Corporation. All rights reserved.

C:\Users\jake> Get-FileHash "C:\Users\jake\OneDrive\Documents\COMP_3010_local\gerv.gun"

orithm       Hash                                                                   Path
------       ----                                                                   ----
256          0931537889C35226D00ED26962ECACB140521394279EB2ADE7E9D2AFCF1A7272       C:\Users\jake\OneDrive\Docume...


C:\Users\jake> Get-FileHash "C:\Users\jake\OneDrive\Documents\COMP_3010_local\wp.exe"

orithm       Hash                                                                   Path
------       ----                                                                   ----
256          79D503165D32176842FE386D96C04FB70F6CE1C8A4858379578492 97E625EA48      C:\Users\jake\OneDrive\Docume...


C:\Users\jake> Get-FileHash "C:\Users\jake\OneDrive\Documents\COMP_3010_local\trwo.exe"
olve-Path : Cannot find path 'C:\Users\jake\OneDrive\Documents\COMP_3010_local\trwo.exe' because it does not exist.

WINDOWS\system32\WindowsPowerShell\v1.0\Modules\Microsoft.PowerShell.Utility\Microsoft.PowerShell.Utility.psm1:110
r:36
                $pathsToProcess += Resolve-Path $Path | Foreach-Objec ...
                                   ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
 + CategoryInfo          : ObjectNotFound: (C:\Users\jake\O..._local\trwo.exe:String) [Resolve-Path], ItemNotFoundE
xception
 + FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.ResolvePathCommand

C:\Users\jake> Get-FileHash "C:\Users\jake\OneDrive\Documents\COMP_3010_local\trow.exe"

orithm       Hash                                                                   Path
------       ----                                                                   ----
256          94A0A09EE6A21526AC34D41EABF4BA603E9A30C26E6A1DC072FF45749DFB1FE1       C:\Users\jake\OneDrive\Docume...


C:\Users\jake>
```

The JS confusion

7497 - when searching for JS, this email pops up
11515 - has commands in the mail

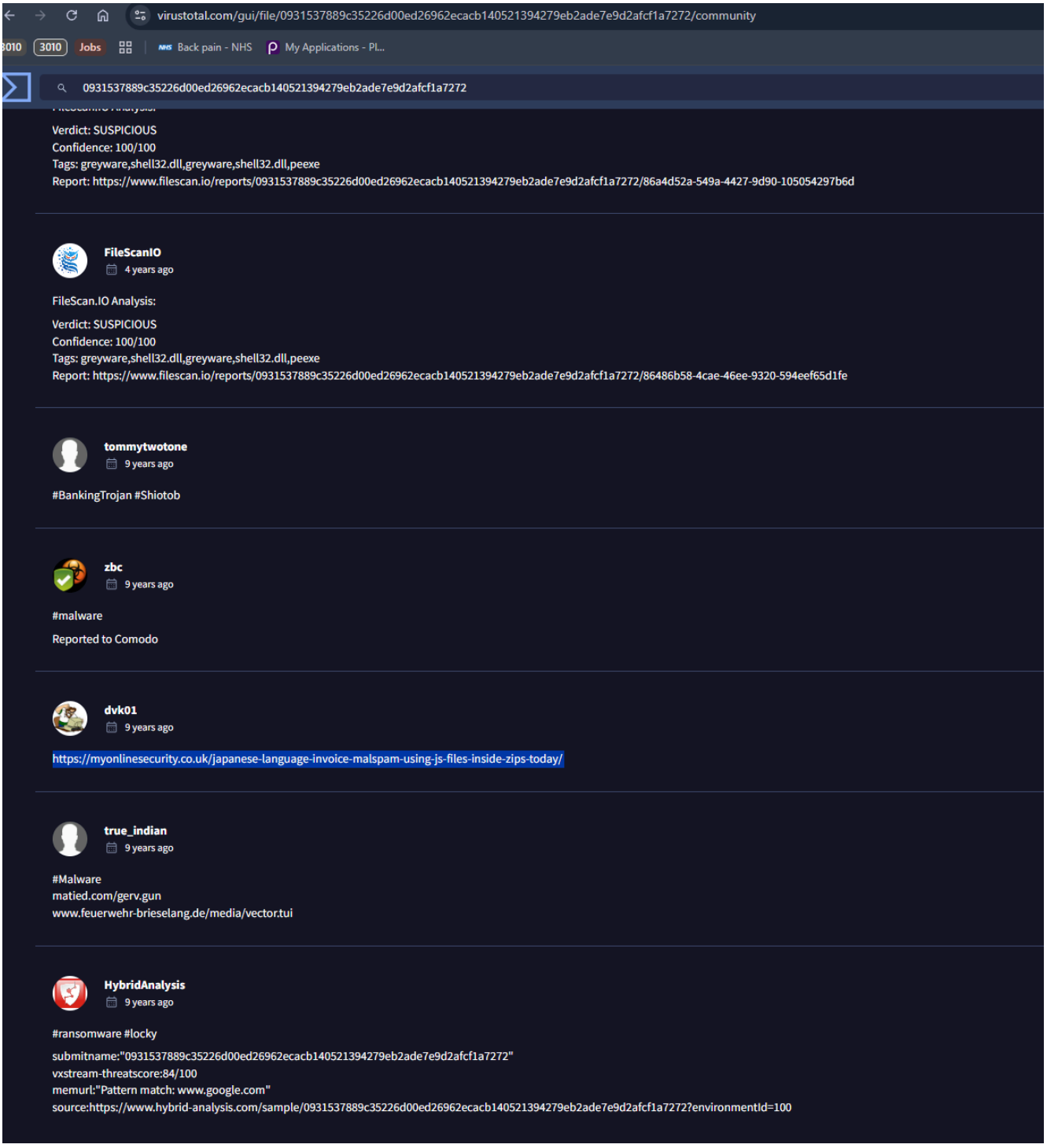
10944 is a very suspicious email and following the SMTP data related to the infected machine, there could be a lead.

[**] [1:23832:4] INDICATOR-OBFUSCATION non-alphanumeric javascript detected [**]
[Classification: Attempted User Privilege Gain] [Priority: 1]
06/27-13:44:01.879024 104.28.18.104:80 -> 192.168.1.96:49210
TCP TTL:53 TOS:0x0 ID:18289 IpLen:20 DgmLen:4971 DF
*****AP*** Seq: 0x48CA8EF8 Ack: 0x36378A6A Win: 0x1F TcpLen: 20**
[Xref => http://patriciopalladino.com/blog/2012/08/09/non-alphanumeric-javascript.html]
^ This references JS
6packet 1738

[**] [1:35180:1] POLICY-OTHER Remote non-JavaScript file found in script tag src attribute [**]
[Classification: Potential Corporate Privacy Violation] [Priority: 1]
06/27-13:44:02.236717 184.168.47.225:80 -> 192.168.1.96:49201
TCP TTL:49 TOS:0x0 ID:31959 IpLen:20 DgmLen:828 DF
A* Seq: 0x4FBCB4B8 Ack: 0xD5E42858 Win: 0x80 TcpLen: 20
[Xref => http://technet.microsoft.com/en-us/security/bulletin/MS15-065][Xref =>
http://cve.mitre.org/cgi-bin/cvename.cgi?name=2015-1729]
^1966
Packet 204 is the response to the gerv.gun request

- I cant find any information on the wire shark so using external tools:

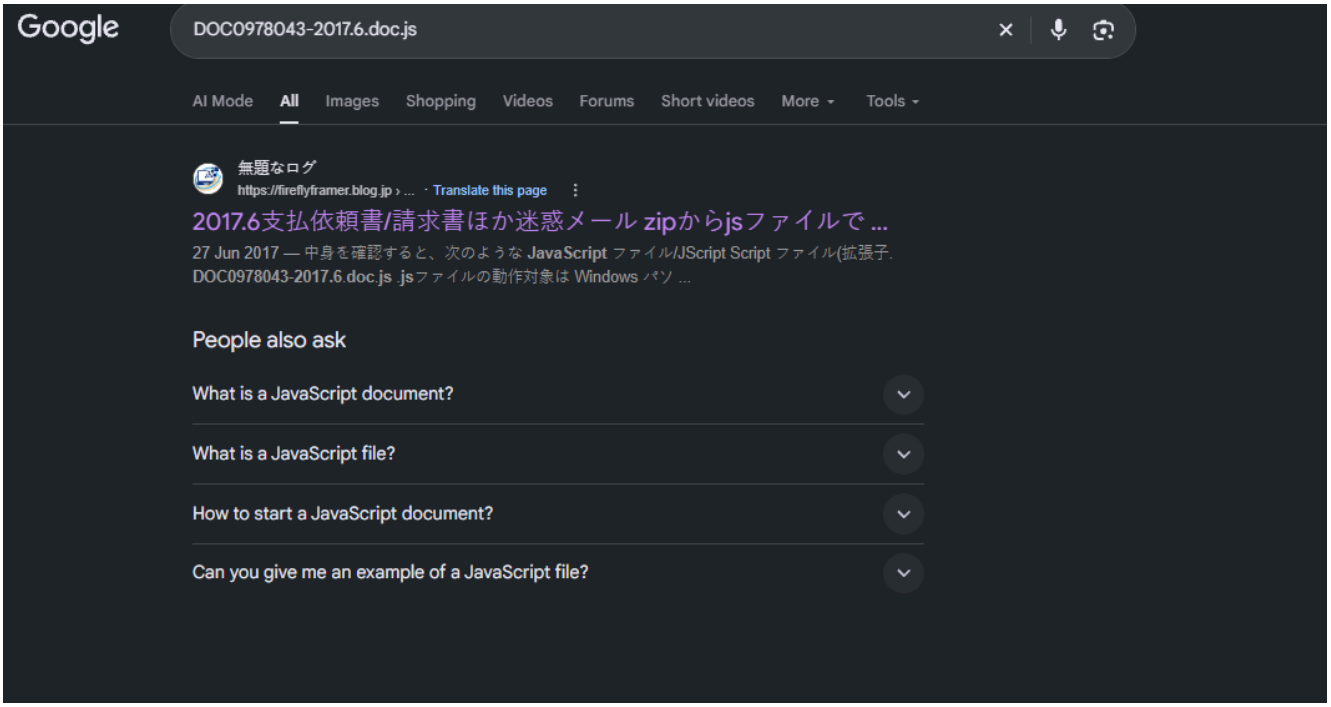
Using virusTotal, I put in the sha256 of gerv.gun:
https://www.virustotal.com/gui/file/0931537889c35226d00ed26962ecacb140521394279eb2ade7e9d2afcf1a7272
0931537889c35226d00ed26962ecacb140521394279eb2ade7e9d2afcf1a7272
I couldn't understand much of the information, however in the comments I found a link, that mentioned JS files in zips, malspam and Japan, which looked like it was the same scenario



The page:
https://myonlinesecurity.co.uk/japanese-language-invoice-malspam-using-js-files-inside-zips-today/

Describes mass mall-spam with emails containing javascript files disguised as invoices, when extracted, the files act as malware downloaders , this sounds like the JS file that was likely used here The page describes the exact issues that are present in the PCAP like matied.com downloading gerv.gun file, and this comes from a JS file called “DOC0978043-2017.6[.]doc[.]js”

Taking the assumption that this is the correct file, as it does exactly what the malware that infected the system in the coursework PCAP, I searched this file, there was one result the came up, from a japanese website post dated to 27-06-2017 (the exact same date as the PCAP). Seeing the familiarity, I followed this site: “https://fireflyframer.blog.jp/19064394.html”



When translating the page, it's clear that this is a blog post titled “June 2017: Virus infection via JS file in spam email containing payment request/invoice and other documents (ZIP file).” where it describes the same situation as the blog post before, with emails that mislead users into downloading a ZIP file full of javascript files, three of them.

The blog outlines that this is a Shiotob virus, which matches with the comments on gerv.gun’s virus total analysis (where I found the link to the original blog).

It outlines three scripts, their MD5 and the virus total link:

> [Script Files]
> ■ MD5 0994c2e3dc7d38b9d6abba5d9755b819
> www.virustotal.com/ja/file/60834d0c9d0602ef18cd65289b0b0fab05bbdb512acd3af0b273b6414883f60e/analysis/1498524960/
>
> ■ MD5 6d8f83550b5f4fea50378ec4bd01282b
> www.virustotal.com/ja/file/59727ea201d94d471bdbb353b1281cca3c5bebf5305b140a3c017e52a37be702/analysis/1498510729/
>
> ■ MD5 0855b3cc00cf86a71d5b1ae31a57bdd2
> www.virustotal.com/ja/file/64ae13592c5d8b289be81ebb50e054cef49b0bdf50ea92dd44ed611dd274150f/analysis/1498523440/

When visiting the VirusTotal analysis’ of each of these, I went to the details section and scrolled down to “names”. The name of the third analysis in the list of sha-256 hash: 64ae13592c5d8b289be81ebb50e054cef49b0bdf50ea92dd44ed611dd274150f has the name of “DOC0978043-2017.6[.]doc[.]js” - this name matches the file name, stated in blog 2, that extracts the gerv.gun. The first blog also states that a different zip file extracts 2017-547805549-231[.]pdf[.]js which contacts the same sites and downloads the same malware as the other js file, and its SHA code is: 59727ea201d94d471bdbb353b1281cca3c5bebf5305b140a3c017e52a37be702.

Despite the quiz stating that both of these two hashes of their respective JS files are incorrect, I believe that I have provided sufficient evidence to show that these are likely contenders to being the culprit JS files that "*PROBABLY*" infected the computer

```
[**] [1:11192:20] FILE-EXECUTABLE download of executable content [**]
[Classification: Potential Corporate Privacy Violation] [Priority: 1]
06/27-13:38:34.111294 119.28.70.207:80 -> 192.168.1.96:49184
TCP TTL:128 TOS:0x0 ID:648 IpLen:20 DgmLen:19976 DF
***A**** Seq: 0xEEED7BAB  Ack: 0xA337D497  Win: 0x500  TcpLen: 20
[Xref =>
http://www.microsoft.com/smallbusiness/resources/technology/security/practice_safe_computing_and_thwart_online_thugs.mspx]

[**] [1:23832:4] INDICATOR-OBFUSCATION non-alphanumeric javascript detected
[**]
[Classification: Attempted User Privilege Gain] [Priority: 1]
06/27-13:44:01.879024 104.28.18.104:80 -> 192.168.1.96:49210
TCP TTL:53 TOS:0x0 ID:18289 IpLen:20 DgmLen:4971 DF
***AP*** Seq: 0x48CA8EF8  Ack: 0x36378A6A  Win: 0x1F  TcpLen: 20
[Xref =>
http://patriciopalladino.com/blog/2012/08/09/non-alphanumeric-javascript.html]
```

The final Stage of analysis consisted of locating the likely point of origin of the JS file that caused the attack. Many Hours were spent dissecting wireshark, Snort and Suricata logs, however there was no evidence of a JS file. During analysis of the gerv.gun VirusTotal summary, there was a comment to https://myonlinesecurity.co.uk/japanese-language-invoice-malspam-using-js-files-inside-zips-today/.